



How do decision makers' ethnicity and religion influence the use of forests? Evidence from Chile

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ABSTRACT

Increasingly it is recognized that traditional models of rational decision making fail to capture important aspects that extend beyond just assessments of relative costs and benefits, expanding to consideration of other elements such as culture. We studied how ethnicity and religion may influence the use of native forests in the South of Chile, an area of high biological and cultural diversity where individual land tenure predominates. A sample of 275 forest owners were surveyed (social and environmental data), of which 173 participated in economic games to measure both risk aversion and time discounting rates. Risk and time behaviors showed significant variations among ethnic and religion groups that relate in turn to different intensities of timber extraction and tree plantations cover. However, we did not find a direct relationship between ethnic and religious groups and the use of forests; instead, we found an indirect influence about how they shape attitudes towards time and risk. The variable for ethnicity in our study was less a measure of ethnicity itself as it is a measure of ethnically based discrimination that has taken place over time. In order to be more effective, Chilean forest policy should focus on reducing uncertainty about forest management decisions, because uncertainty exacerbates negative behavior about forests by increasing risk aversion (more intense logging) and time discounting rates (less investment in tree plantations), especially in most critical socioeconomic and environmental contexts. This would help decision makers move towards more sustainable practices.

1. Introduction

Despite recent assessments observed that forests¹ are recovering in some countries and specific territories, deforestation and forest degradation remain a global problem (FAO, 2016). In order to sustain forest management and conservation, there has been a focus to understand why destructive processes are happening. Social issues are usually on the list of drivers, namely poverty and other indicators of poor socioeconomic conditions, but lately cultural aspects have also emerged as important factors to take into account (Stevens et al., 2014; Vergara and Potvin, 2014; Walker et al., 2014).

Forests are one of the natural resources used by landowners to satisfy their needs. They are part of productive systems that sustain people's livelihood, so landowners' decisions, productive systems, and the state of forests are closely linked, especially where private land tenancy predominates (Henrich, 2002). This is the case of Chile where 70% of

forests are privately owned (de la Fuente et al., 2013). Forest owners' decisions are influenced by their socioeconomic and biophysical context (Amacher et al., 1996; Heltberg et al., 2000; Heltberg, 2002), but they also depend on psychological aspects (Henrich, 2000, 2016). The way as those, represented in this research by risk aversion and time discounting rate, relate to decision-makers' cultural origin (ethnicity and religion) and, in turn, to forest management decisions, is the research gap that this article aims to contribute to fill.

The psychological dimension of decision-making is somewhat a cultural construction, as it is influenced by cognitive biases (Kahneman and Tversky, 1979) that every person develops along their childhood and adolescence. Aspects like ethnicity, religion, and beliefs would determine, among other things, a person's tolerance to uncertainty (Henrich et al., 2001; Türetgen et al., 2008; Henrich et al., 2010). This, in turn, could produce different behaviors concerning risk aversion and time discounting, which can become key aspects about the use of forests

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¹ In this research, the concept of forest is synonymous of native forests, while forest plantations (mainly fast-growth Eucalyptus and Pines plantations in Chile) are referred as tree plantation.

(Andersson and Gong, 2010).

While controlling by socioeconomic and biophysical variables, this article aims to analyze the influence of decision makers' ethnicity and religion on the use of forests (timber extraction rate and tree plantation cover), which is relevant in a culturally diverse context like the Chilean one. For that, we assess if ethnicity and religion are related to differences in time discounting rate and risk aversion, and if those influence behaviors on how forests are being managed. Socioeconomic and biophysical variables are adjusted to see if ethnicity and religion influence behaviors beyond their joint effect, as risk aversion and time discounting rate are correlated to income, schooling and other contextual factors (Yesuf and Bluffstone, 2009; Dohmen et al., 2010; Tanaka et al., 2010). Then, we look at timber extraction rates (consumption of forests) and tree plantation cover (investment in new forest) to see if ethnicity and religion directly relate to the use of forests, or if this relationship is more indirect. Finally, we offer some thoughts as to how these aspects could influence land use, and some policy recommendations to improve the effectiveness of forest regulations.

1.1. Forest regulations in a culturally diverse context

In Chile, manageable forests are in hands of thousands of individual landowners along the country, as communitarian land tenure is not common and most of state-owned forests are protected (i.e. national parks) (Reyes and Nelson, 2014). According to the National Forest Inventory, 77% of forest owners have formal land tenure (legalized rights) and the remaining 23% are in some informality status (Reyes, 2020). Regularized forest owners can ask for a legal authorization to harvest timber through a Forest Management Plan (FMP), which has to be previously approved by the National Forest Service (CONAF). This authorization lasts some years, and contains a detailed list of activities that forest owners must perform during the period. However, all the administrative process and its later implementation is full of barriers and uncertainty, which leads forest regulations are poorly implemented, illegal logging is frequent, and forests end up degraded (Lara et al., 2016).

Forest regulations are difficult to understand due to its high technical and administrative complexity, so landowners usually harvest their forests without understanding what the National Forest Service approved for them (de la Fuente et al., 2013). Additionally, landowners have to compete to get public funding to implement good practices in their forests, which implies costs as landowners have to hire a Forest Engineer to complete the application, and when this is successful, money is paid between two and five years after silvicultural activities are implemented (de la Fuente et al., 2013). This implies people have to spend their own money along the process and be patient in order to recover the investment, as well as being risk tolerant as money is not reimbursed if silvicultural activities goes wrong (de la Fuente et al., 2013). Forest regulations assume all landowners assess uncertainty similarly.

Consequently, forests (native forests) are just used to supply a highly informal firewood market (firewood for heating in rural and urban areas, which accounts for about 9 million of cubic meters per year), being totally replaced by industrial tree plantations (*Eucalyptus* sp. and *Pinus radiata*) as a timber supplier for other markets (international) and products (pulp, construction materials, chips, etc.). Regulatory uncertainty increases when ecological complexity of forests is taken into account, as they need more time to grow and their payoffs can be heavily discounted. Forest owners have to be patient and risk tolerant, but it is not clear how different kind of decision makers face this "lack of knowledge" about future outputs and consequences.

The importance of studying this cultural dimension about the use of forests was suggested by de la Fuente et al. (2013) in the last assessment of the implementation of the Native Forest Law. They identified "cultural barriers as underlying factors that undermine effectiveness of forest regulations in Chile". In that context, how do aspects related to

culture, like ethnicity and religion, and the socioeconomic and biophysical context of decisions makers influence the way they face uncertainty and use forests in Chile? Opening this black box, which has not been addressed in previous studies, would contribute to increase the effectiveness of Chilean forest policy.

1.2. The historical context

In pre-Columbian times, several first nations lived in what is the Chilean territory today. One of these nations, named Mapuche ("people of the land" in *mapuzungun*, the Mapuche language), was a vigorous nation comprising about one million people when Spanish conquerers arrived to this latitude (Bengoa, 1985). Mapuches and Spaniards had a long war, which finished in 1641 when the Spanish Empire recognized Mapuche sovereignty in a vast territory from the Biobío River to the south (this includes the Los Ríos region) (Bengoa, 1985). Mapuche was the only first nation along Americas that was never defeated by the Spanish Empire, which was recognized and respected by the Spanish Crown during the colonial period (Bengoa, 1985).

Chile was born as an independent republic in 1810 and, in 1880, decided to invade the Mapuche territory in a military operation known as the "Pacificación de la Araucanía" (Pacification of Araucanía²). The Mapuche people did not have an "army" nor a unified and independent state, at least not how it is considered by the western culture, so this military operation was not oriented to fight against another army, but basically expelling Mapuche people from their own lands (Bengoa, 1985). This ended in an abusive and infamous slaughter by the Chilean army. The surviving Mapuche families were confined to indigenous reservations. Later, the Chilean government started to implement a colonization policy, which facilitated people from the north of the country and more recent European immigrants (mainly German, Swiss and Italian settlers) to get legal rights over lands that previously were Mapuche. This policy considered economic incentives to crop the land, which, among other impacts, increased wildfires and deforestation (Camus, 2003).

In the last few decades, the Chilean State has been implemented a policy to give lands back to Mapuche families. This has improved the socioeconomic conditions of the Mapuche population while, at the same time, decreasing the physical segregation between ethnic groups in rural areas. However, from a socioeconomic perspective, Mapuche landowners tend to be poorer than non-Mapuche ones and own smaller pieces of land. Indeed, significant heterogeneity among ethnic groups has been found (Agostini et al., 2010). Unlike Mapuche, European descendants normally own large plots, which are located in the best agricultural soils. This also translates into higher schooling, more access to credits, and higher political influence than the rest of landowners. Chilean mestizos are, in turn, the largest group, and show a significant variability in wealth, education, plot size, and other aspects, having a large percentage of them at a similar socioeconomic condition as Mapuche landowners. It should be noted that while significant disparities exist on average across ethnic groups, only 4.4% of rural population lives in extreme poverty (Ministerio de Desarrollo Social, 2017).

Along time a "chilenization" process, strongly driven by outside religion and forced schooling, has affected to the Mapuche culture (Green, 2020). Today, being ethnically Mapuche does not necessarily mean to ascribe to the same worldview and customs. For this research, this implies that ethnicity, while it is related to cultural belonging, does not perfectly equate with culture in the wider understanding of the concept. While this is one of the limitations of this research, it also means the results can be considered "conservative" in that they include responses under the Mapuche category by people with varying degrees

² Arauco is the name used by the Spanish conquerers to refer to the Mapuche territory, and it means "clay water" in *mapudungun* (Mapuche language) or "wild/indomitable" in Quechua (Inca language) (Bengoa, 1985).

of adherence to traditional approaches to the forest.

2. Theoretical background

Decision-making is not undertaken by only analyzing the costs and benefits of different choices, it is rather a biased process that strongly depends on the decision maker's characteristics and experiences (Henrich, 2002), and his/her socioeconomic and ecological context (Amacher et al., 1996; Heltberg et al., 2000; Heltberg, 2002; Joshi and Mehmood, 2011). The influence of the socioeconomic context on people's decisions has been extensively studied, and typically focuses on factors such as income, education, and age as the most relevant aspects that affect decision-making. (Frederick et al., 2002; Ventura, 2003). However, deeper in the human being, in a more psychological dimension, aspects like uncertainty, risk aversion and time discounting have emerged as underlying drivers of people's behavior over the last few decades (Henrich et al., 2001).

The decision of managing forests represents a social dilemma as forest owners' interest can clash against the society's interest. According to Dawes (1980), this happens when the personal benefit for acting in self-interest (for instance, unsustainable logging) is higher than the personal benefit for acting in the interest of the society (taking care of forests to provide ecosystem services), although both forest owner and the rest of the society benefit less when everybody decides to defect rather than cooperate. Weber et al. (2004) point out that people face social dilemmas by asking themselves "What does a person like me do in a situation like this?" Cooperative behaviors, like managing forests sustainably, are affected by uncertainty as decision makers ignore how others could act, and the consequences of their decisions on the environment (Biel and Gärling, 1995).

Uncertainty, understood as "lack of knowledge that results from intrinsic variability in the system(s) or processes under consideration" (Ascough et al., 2008), is a key piece in the research about the use of natural resources. Several sources of uncertainty interact through the decision making process: environmental variation (system's own variability), partial controllability (imprecision attributable to management actions), structural uncertainty (non-expected consequences or results), and partial observability (this recognizes that decision-makers rarely observe all components of nature) (Nichols et al., 2011). Relatively new phenomenon, like climate change, are increasing the variability and unpredictability of these uncertainty sources, which makes more relevant to include this aspect in research about forest policy (Polasky et al., 2011).

In presence of uncertainty, people are selective on what information they consider it is useful, and try to find shortcuts to analyze it before taking decisions (Weber and Johnson, 2009). Decision makers are motivated to reduce "what they cannot completely understand, predict and control", which is interpreted as an aversive state (Bar-Anan et al., 2009). The way as people face risk is relevant in terms of the final answer in presence of a social dilemma, as also how they lead with the time dilemma that decisions about managing forests can imply (Korntenkamp and Moore, 2006). According to these authors, cooperative behaviors are more common when the temporal dilemma is reduced, as also the risks about the environmental consequence of people's decisions. Javald (2016) mentions that people opt for more destructive harvesting methods when benefits of conservation are delayed.

Forest owners' risk aversion and time discounting can change depending on aspects like ethnicity and religion (Fig. 1), which have been increasingly recognized as an important component of economic behavior (Henrich, 2000, 2016; Henrich et al., 2001). For instance, in Chile, some authors note that landscapes where Mapuche decision makers predominate have higher forest cover (Montalba and Carrasco, 2003; Fatheuer, 2011; Montalba and Stephens, 2014). This is generally attributed to Mapuche people's traditional ecological knowledge and worldview, as forests have a dense symbolic meaning for the Mapuche due to they consider everything is deeply connected and has a physical

and spiritual dimension (The *Ngen*, or spirit in *mapuzungun*) (Herrmann and Torri, 2009). However, it is not clear if ethnicity, religion or other cultural characteristics influence the way as people face social dilemmas by themselves, or they interact with the socioeconomic context where these decisions are taken.

Due to the high level of income and land access inequality between Mapuche and non-Mapuche forest owners, it is important to assess how ethnicity and religion could be driving forest owners' decisions for their own, beyond poverty and other socioeconomic drivers (Shah et al., 2012). For this, the main socioeconomic factors that influence risk aversion and time discounting were identified and adjusted to isolate the effect of ethnicity and religion, and assessing their own influence on the use of forests. We hypothesize that beyond the socioeconomic context, both ethnicity and religion would influence the use of forests (timber extraction and forest cover) by changing the way as forest owners face uncertainty (through risk aversion and time discounting).

3. Methods

3.1. Study area

The Los Ríos Region is located between 39°15' and 40°33' south latitude in south-central Chile, the heart of the Valdivian temperate rainforests. It covers 18,400 km² and is divided into three physiographic units: the Coastal Range, the Central Valley, and the Andes Range (Fig. 2). The Region is bordered by the Araucanía Region to the north, the Los Lagos Region to the south, Argentina to the east, and the Pacific Ocean in the west. This area has a temperate oceanic climate (Cfb) with an average rainfall of 2100 mm per year and a mean temperature of 12.9 °C (Castillo, 2001).

Forests cover 47% of the area of the region followed by 30% grasslands and shrublands, 11% tree plantations (*Pinus radiata* and *Eucalyptus* sp), 6% lakes and rivers, and the remainder (6%) comprising other land uses (CONAF, 2008). Forests show significant differences along the edaphoclimatic gradient; on the coast, evergreen forests predominate, while deciduous forests (*Nothofagus* sp.) are most common in the Central Valley and the Andean Range (Donoso, 1993). All these forests are dominated by broadleaved tree species.

The main economic activities in the Los Ríos Region are agriculture, livestock and dairy production, and tourism. Of the 384,837 people in the region, 72% live in urban areas. Valdivia city is the capital with 150,000 inhabitants followed by La Unión (26,000), Panguipulli (17,000) and nine other smaller cities (INE, 2017). Mapuche (first nation) make up 26% of the population, 45% follow Catholicism, and 11% are evangelical Christians (INE, 2017). About 12% of households are considered income poor, and income averages about US\$1000 per household per month (Ministerio de Desarrollo Social, 2017). However, the socioeconomic condition of families is very unequal, with Mapuche people being poorer than other ethnic groups, having a higher unemployment rate, and living on smaller pieces of land (Celis et al., 2008).

3.2. A model of decision making

The Los Ríos Region provides a good test-bed to analyze the influence of decision makers' ethnicity and religion on the use of forests, due to the high variability of decision makers in social and cultural terms. For this analysis, ethnicity was represented by three groups: *Mapuche* (first nation), *Mestizo*,³ and post-colonization European descendants,⁴ all having Chilean citizenship, and religion by two groups: Catholic and Evangelical. Although these categories are not rigid, and there are

³ Mixture of indigenous and European people with more than two generations in Chile.

⁴ European descendant, mainly German and Swiss, until the second generation.

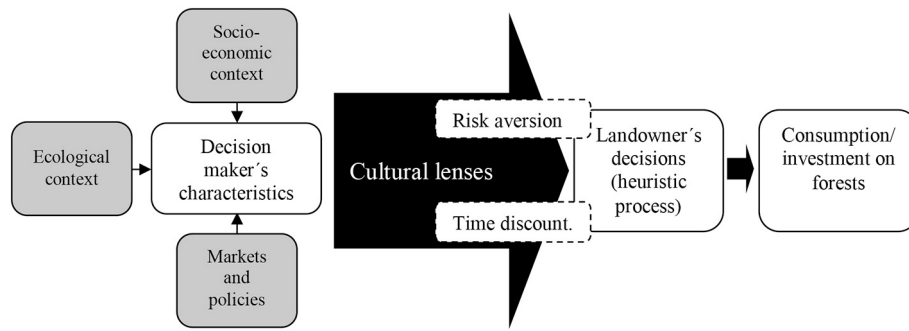


Fig. 1. Landowner's decision making process concerning forests.

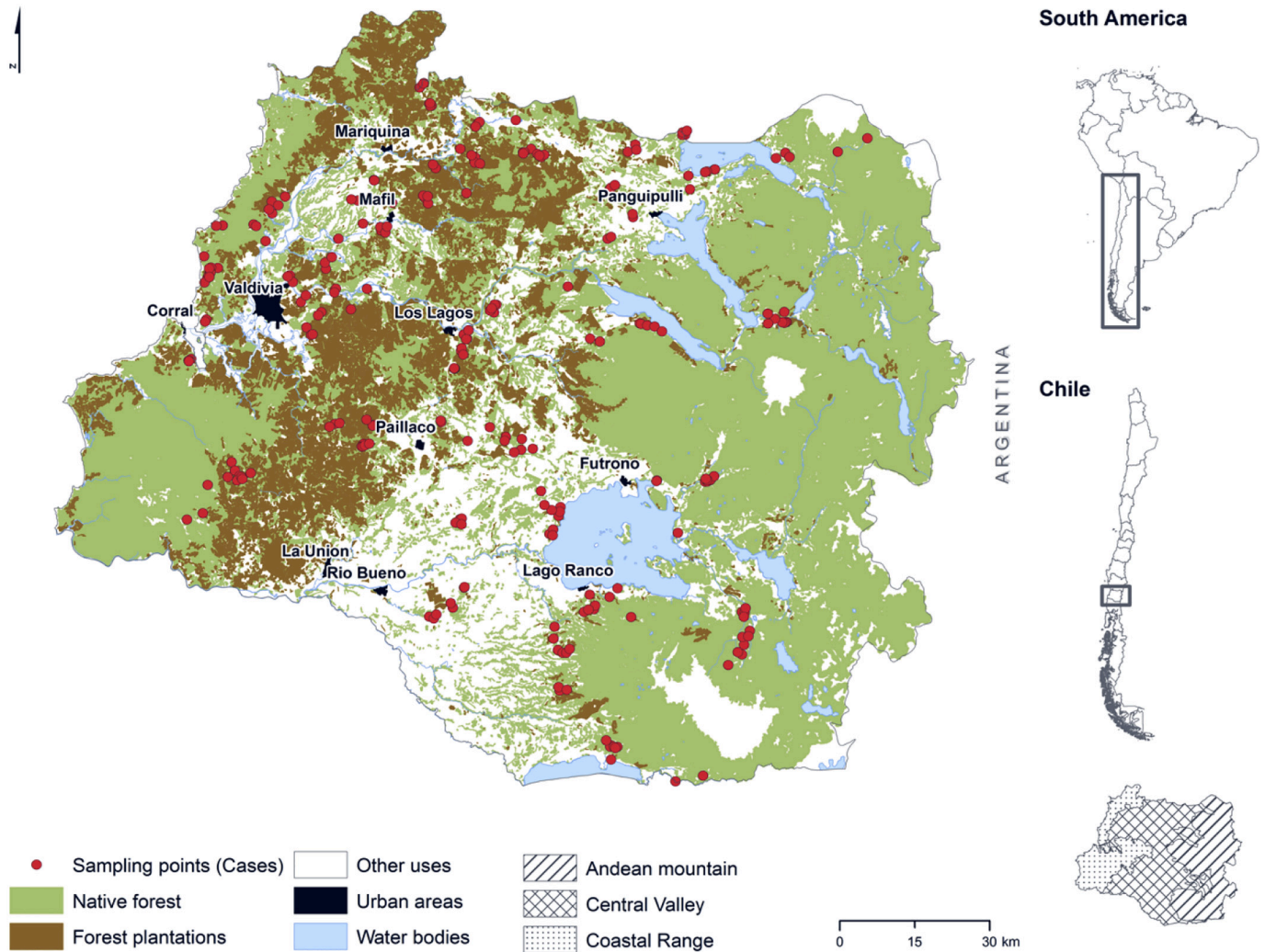


Fig. 2. Study area (Los Rios Region) and sampling points. Urban areas correspond to the capital cities of the 12 counties in the region.

different levels of “ethnicity” or “religiousness”, discrete classifications were used to reduce subjectivity based on the survey and the individual judgment of participants.

To analyze the relationship between the use of forests (measured as both consumption of forests and investment in new forests), the decision makers' socioeconomic and biophysical context, and ethnicity and religion, which would be hypothetically channeled through their preferences concerning risk and time, the following models were tested:

$$R_i = f(C_i, X_i, V_i, U_i, E_i) \quad (1)$$

$$T_i = f(C_i, X_i, V_i, U_i, E_i) \quad (2)$$

Where,

R_i = risk aversion, decision maker i .

T_i = time discounting, decision maker i .

C_i = ethnicity and religion, decision maker i .

X_i = decision maker and decision maker's family characteristics, decision maker i .

V_i = economic and ecological vulnerability, decision maker i .

U_i = dummy variable for trust, decision maker i .

$\mathcal{E}_i$ = other non-considered characteristics of decision maker i .

Time discounting was measured as a monthly rate that expresses the preference for present rewards instead of future rewards (discounting rate). In turn, risk aversion was measured as a rate that expresses the relationship between the amount of money that a person is willing to lose to avoid risk, and the money that a risk-neutral person would earn in the same situation. Therefore, larger positive numbers represent greater risk aversion.

Before starting the time discounting and risk aversion experiments, each decision maker was asked to answer the following “question of trust”: *Generally speaking, would you say that your neighbors are trustworthy?* Answers were used as an adjustment variable in statistical analyses to see the potential effect of trust perception on time and risk preferences (see eqs. 1 and 2). In experimental economics, the question of trust is part of the normal procedure to analyze human behavior. This allows the relative perception of trust of participants to be included in the analysis (Karlan, 2005).

Additionally, the use of forests was also tested, starting with the intensity of timber extraction (strategy 1: forests as a checking account), followed by the relative abundance of tree plantations (strategy 2: trees as an investment or saving account). Both strategies influence the current and future state of forests, as they are generally reduced in area or cover by logging, and increased by tree plantations.

The following models were tested:

$$FUC_i = f(R_i, T_i, X_i, Z_i, V_i, U_i, \mathcal{E}_i) \quad (3)$$

$$FUI_i = f(R_i, T_i, X_i, Z_i, V_i, U_i, \mathcal{E}_i) \quad (4)$$

Where,

FUC_i = consumption of forests (harvesting intensity in $m^3/ha/year$), decision maker i .

FUI_i = investments (tree plantation cover in percentage), decision maker i .

R_i = risk aversion, decision maker i .

T_i = time discounting, decision maker i .

X_i = decision maker and decision maker's family characteristics, decision maker i .

Z_i = characteristics of the productive system, decision maker i .

V_i = economic and ecological vulnerability, decision maker i .

U_i = dummy variable for trust, decision maker i .

$\mathcal{E}_i$ = other non-considered characteristics of decision maker i .

Table 1 shows the variables used in the analysis.

3.3. Field methods

A cross-sectional sampling was performed in 2013.⁵ The Los Ríos region is divided into twelve counties that, in turn, are broken into 100 census districts.⁶ Of these, 30% were randomly chosen from each county. Three points were randomly marked inside each selected census district using the “randomize” application in ArcView 3.2, and satellite images available in Google Earth were used to identify the sampling points in the field.

Once in the field, the closest four plots to each sampling point were surveyed. Plots were defined as the sum of all pieces of land owned/used by the same decision maker in the Los Ríos Region, with presence of forests. When the main decision maker⁷ was not available at the plot,

⁵ A pre-sampling was performed in 2012 to identify decision makers, introduce the research, and generate trust for the main sampling in 2013. The respondent was the same person for both visits.

⁶ To implement the national census, the National Institute of Statistics divides each county into areas according to their populations, and these areas are termed Census Districts.

⁷ This was defined as the person in charge of the plot, normally the land-owner or the manager.

Table 1
Variables.

Group	Variable
Decision maker's preferences	Time discounting (%) ^a Risk aversion ^b
Decision maker and decision maker's family characteristics	Ethnicity (<i>Mapuche</i> , Mestizo, and Post-colonization European des.) Religion (Catholic or Evangelical) Sex (man or woman) Age (years) Education (years) Family size (quantity) Forest cover (%) Forest area (hectares) Share of domestic consumption (%) ^c Share of off-farm incomes (%) Quantity of cattle Land tenure (formal or informal)
Characteristics of the productive system	Total income (million Chilean pesos per year) Plot size (hectares) Distance to the closest town (km) Distance to the largest city in the region (km) Forest growth (cubic meters per hectare per year) Plot size adjusted by quality of land
Economic and ecological vulnerability	

^a Monthly interest rate (in percentage).

^b Difference between the last safe option taken by participants before betting and a risk-neutral decision maker (participant is indifferent between betting and not betting when the safe option is the same as the middle point between both sides of the coin). Therefore, larger positive numbers represent greater risk aversion.

^c Proportion out of a total of eight products that are produced and consumed on-farm, including milk, potatoes, vegetables, fruits, charcoal, firewood, meat and eggs. This represents the level of autarky of the productive system (self-sufficiency).

his/her cellphone number was obtained from neighbors, and the survey was performed later. Informed consent was obtained from decision makers using a research protocol approved by the behavioral-research ethics board of the University of British Columbia. If he/she did not want to participate in the study or could not be found, another plot was chosen as a replacement. In total, 275 decision makers were surveyed, of which 173 participated in the two tests to elicit time discounting and risk aversion. The Fig. 2 shows the location of the 173 decision makers in the Los Ríos Region.

The survey was an improved version of surveys used in previous research related to forest owners, an expert consultation process performed between 2011 and 2012, and the use of the 2007 Agricultural Census (INE, 2007). Income was not directly asked because people rarely reveal this information. Therefore, it was estimated based on on-farm production and off-farm activities. On-farm income comes from agricultural and forestry incomes, while off-farm income comes from pensions and subsidies, salaries, and off-farm production. In the case of agriculture, stockbreeding, forest production, and off-farm production, income is production valuation. In the case of salaries, which considered those of all family members, they were estimated based on regional averages (average payments for specific kinds of jobs).

Similarly, timber production was estimated based on crossed questions related to on-farm production (time and people dedicated to different activities, and productivity values), and considering firewood, charcoal, and roundwood production (the main timber forest products in Chile).

3.4. The measurement of temporal discounting and risk preferences

A set of economic games was carried out with decision makers to measure time discounting and risk aversion by using hypothetical amounts of money, and a reward at the end of the experiment. Although

the use of hypothetical stakes has been considered to be an inaccurate method in experimental economics (Hertwig and Ortmann, 2001; Holt and Laury, 2002), results obtained by using real and hypothetical amounts of money are positively correlated (Ding et al., 2010). Furthermore, even when real payments are used, the literature points out other factors that also produce inaccuracies on the results: personality traits, quantity of trials that are paid, different magnitudes of monetary rewards, among others (Greibitus et al., 2013; Schmidt and Hewig, 2015; Xu et al., 2016). Therefore, the relationships among variables and ethnic and religion groups (trends) would be more important for the analysis than the magnitudes of risk aversion and time discounting themselves.

The first game aimed to estimate time discounting rates based on the multiple price list method (Harrison et al., 2005). Participants were classified into three categories for both experiments, according to their income levels provided by the survey. In the first level, the amounts used in the experiments fluctuated between 200,000 and 400,000 Chilean pesos per month (\$400–800 USD⁸). In the second level, the amounts varied between 1000,000 and 2000,000 Chilean pesos per month (\$2000–4000 USD), and in the third level, the amounts fluctuated between 10,000,000 and 20,000,000 Chilean pesos per month (\$20,000–40,000 USD). The large differences in the amounts used in the experiments represent the large income differences observed among decision makers. These categories were included in the statistical analysis to assess if they had any influence on the results.

The time discounting experiment consisted of asking about the participants' preferences concerning one-month and six-month payments to identify the point at which six-month payments become more attractive than one-month payments (Harrison et al., 2005). At the beginning, the interviewer explained the following: *"In this game, you will choose between receiving a certain amount of money in one month or a larger amount in six months. After the game, you will receive a reward according to your participation."*

Before starting, the researcher described the following context: *"Suppose that you will receive a donation from an institution. The institution issues two checks, which stay in possession of the bank. The first one is for 200,000 Chilean pesos [1,000,000 or 10,000,000 depending on the income category] to be paid in one month, and the second one is for 240,000 Chilean pesos [1,200,000 or 12,000,000 depending on the income category] to be paid in six months. You have to choose between them. Which would you prefer?"*

When the participant preferred the first option, the second question was *"If the second check [six months] increases to 280,000 Chilean pesos [1,400,000 or 14,000,000], which you would prefer?"* When the participant again preferred the one-month option, the following questions increased the six-month option in increments of 40,000 Chilean pesos (200,000 or 2000,000) until he or she switched in favor of the six-month option. When, in contrast, the participant preferred the six-month option when asked the first question, the game ended. The indifference point is the middle point between the last six-month option that was rejected by the participant and the six-month option that was accepted (the middle value in the switching point).⁹ For example, a time discounting rate of 12% implies that people will prefer the one-month option until the six-month option is 60% higher.

The following equation was used to estimate time discounting rate (TDR):

$$\text{TDR (\%)} = \left(\left(\frac{\text{IP}}{\text{OM}} \right)^{1/5} - 1 \right) \times 100 \quad (5)$$

Where,

TDR = time discounting rate (%).

IP = indifference point (\$). This is the middle point between the last six-month option that was rejected by participants (in favor of the one-month option) and the six-month option that was finally accepted.

OM = one-month option (\$).

TDR is a monthly discounting rate (%) estimated at the indifference point, where the perception of benefit of the one-month option is equivalent to the six-month option. This represents the relative weight of the present (present benefits, consumption, etc.) concerning the future. A higher time discounting rate implies a greater preference for the present, i.e. the future has a lower weight in the decisions, while a smaller rate represents the opposite.

The second game aimed to estimate risk preferences based on the titration experiment (Henrich and McElreath, 2002; Harrison et al., 2005). This experiment consisted in asking participants to choose among a set of binary choices that involved certain sums of money and risky bets. The goal was to estimate an indifference point, where people did not show a clear preference between a safe position and the bet. The risky bet was represented by a coin. One side of the coin was labeled with the number 400,000 (2000,000 or 20,000,000 depending on the income category) and the other side with \$0.

Before starting the experiment, the researcher explained the following: *"In this game, you will choose between receiving a certain amount of money [sure money] or betting to get a larger amount. After the game, you will receive an award according to your participation."* The context of the game is this: *"Suppose that you will receive a donation from an institution. The institution gives you two options: the first is receiving 200,000 Chilean pesos now [1000,000 or 10,000,000 depending on the income category], and the second is flipping a coin with the chance of getting 400,000 Chilean pesos [2000,000 or 20,000,000 depending on the income category] or nothing [zero]. Which would you prefer?"*

When the participant preferred the first option, the safe option was reduced by 40,000 Chilean pesos (200,000 or 2000,000, respectively) in the second question. When, in contrast, the participant preferred the risky option in the first question, the following questions increased the safe amount of money by 40,000 Chilean pesos (200,000 or 2000,000, respectively). The game ended when the participant changed their decision from the safe to the risky option or vice versa. The indifference point was the middle value between the safe and the risky options. The following equation was used to estimate the risk aversion rate (RAR):

$$\text{RAR} = \text{RP}/\text{CE}, \text{ and } \text{RP} = (\text{EV} - \text{CE}) \quad (6)$$

Where,

RAR = risk aversion rate.

RP = risk premium (\$). This is the quantity of money sacrificed to avoid betting.

EV = expected value (\$). This represents the middle point between the two values marked on the coin (200,000; 1000,000; or 10,000,000, depending on the income level).

CE = certain equivalent (\$). This represents the last safe option that was chosen before betting.

RAR represents the relationship between the amount of money that participants are willing to sacrifice to avoid betting and the minimum amount of safe money that is of interest to them before betting. Therefore, when the willingness to sacrifice money increases, the risk aversion rate also increases.

3.5. Statistical analysis

The three ethnical groups (Mapuche, Mestizo, and post-colonization European descendants) were compared in terms of different variables by using ANOVA and Tukey's test, in order to identify statistically significant differences among them for later analyses. A similar analysis was performed for the two religious groups by using a *t*-Test.

A Tobit regression analysis was used to identify variables that better explain risk aversion, time discounting, consumption of forests, and

⁸ \$1 USD = 500 Chilean pesos.

⁹ If the participant accepts the six-month option in the first question, the indifference point corresponds to the middle point between the one month option and the six month initial option.

investment on tree plantations (eqs. 1, 2, 3 and 4, respectively), because these variables have a truncated distribution.¹⁰ This method has been widely used in these kind of studies related to behavioral economics (Jiannakoplos and Bernasek, 1998; Dercon and Christiaensen, 2011). This was performed by using the QLIM procedure in SAS®. Due to the large number of potential dependent variables, Pearson coefficients and forward and backward selection methods were used to choose the best variables (iterative process). The Akaike information criterion (AIC) values were used to compare models. All assumptions were checked for normality of residuals and heteroscedasticity of variance, and multicollinearity was assessed using variance inflation factors (VIF).

4. Results

4.1. The decision makers

Out of 173 participants 28% were *Mapuche*, 63% were Mestizos, and 9% were European descendants, while from a religious perspective, 62% were Catholic, 26% were Evangelicals, and the remaining 12% expressed other affiliations. This last group was equally distributed across the three ethnic groups, representing between 12% and 14% of the members in each group. Their main characteristics are presented in Tables 2 and 3.

Table 2 shows statistically significant differences between the three ethnical groups. Forty-one percent of the *Mapuche* decision makers were Evangelical, which is higher than the other two ethnic groups (Mestizo and European descendant decision makers, with 21% and 13% respectively). *Mapuche* decision makers also had a lower schooling.

Livestock and agriculture were the main income sources for European descendants, averaging 84% of the total income. In the case of *Mapuche* and Mestizo, off-farm income is the most important income source, with 59% and 50% of the total income, respectively. Forest production is less important for the three ethnic groups, although there are some plots where forestry can be predominant (>50% of the total income).

European-descendant decision makers have the highest total income. They also have a higher proportion of formal land tenure and larger plots than *Mapuche* and Mestizo decision makers, although in the case of Mestizo decision makers, the plot size shows high variability. Yet, in terms of forest cover there were no statistically significant differences among the three ethnic groups. Therefore, while *Mapuche* decision makers have lower incomes and smaller plots, their forests cover a similar share of the plot than other decision makers.

Tree plantations are more frequent in plots owned by European descendant decision makers (73% of the cases) and cover on average 7% of the piece of land. In plots owned by *Mapuche* and Mestizo decision makers, tree plantations are present in half of the cases, covering about 5% of the land.

Fig. 3 shows a histogram of the risk aversion rates for the three ethnic groups. Larger numbers represent greater risk aversion. *Mapuche* decision makers show the highest risk aversion due to the presence of a large number of people with “extreme risk aversion”. This represents people who always prefer safe money instead of betting. Extreme risk aversion was also found by Wik et al. (2004) and Yesuf and Bluffstone (2009) when the stake amount was significant, as is the case in this study.

In the case of time discounting, Mestizo decision makers have a higher time discounting rate (more impatient) than European descendants (Fig. 4), while that of *Mapuche* decision makers are not statistically different from those of the other two groups.

Moreover, Table 3 shows statistically significant differences between

Table 2

ANOVA test for the main characteristics of the three ethnic groups.

Characteristic	<i>Mapuche</i> (n = 49)		Mestizo (n = 109)		European (n = 15)	
	Mean	Std. dev.	Mean	Std. dev.	Mean	Std. dev.
Religion						
Catholic (%)	45	NA	68	NA	73	NA
Evangelic (%)	41	NA	21	NA	13	NA
Decision maker's age (years)	61 (ab)	12	63(a)	13	55(b)	9
Education (years)	3(a)	2.9	6(b)	5.4	16(c)	2.0
Family size (quantity)	3.7 (a)	1.8	3.1 (a)	1.3	3.4 (a)	0.9
Distance to the closest town (km)	20(a)	11	19(a)	12	19(a)	12
Distance to the largest city in the region (km)	93(a)	35	64(b)	30	64(b)	32
Share of agricultural incomes (%)	28(a)	19	43(b)	32	84(c)	21
Share of forestry incomes (%)	12(a)	18	13(a)	21	11(a)	19
Share of off-farm incomes (%)	59(a)	25	50(a)	29	16(b)	14
Total income (million Ch\$/year)	5.1(a)	2.1	58(a)	174	361 (b)	431
Formal land tenure (%)	66	NA	73	NA	100	NA
Plot size (hectares)	33(a)	48	215 (a)	665	882 (b)	1054
Forest cover (%)	38(a)	26	38(a)	28	35(a)	28
Tree plantation cover (%)	4.8(a)	8.8	4.6 (a)	8.4	7.0 (b)	7.0
Forest growth (m ³ /ha/year) ^a	4.3(a)	2.7	5.2 (a)	3.0	7.7 (b)	2.4
Time discounting rate (%) ^b	5.8 (ab)	4.7	6.8 (a)	6.4	3.2 (b)	2.1
Risk aversion rate ^c	11.5 (a)	8.6	7.3 (b)	7.8	5.3 (b)	6.9
Trust question (% of people who trust)	75	NA	58	NA	75	NA

Note: Letters (a, b and c) represent groups with statistically significant differences (95% confidence) according to ANOVA and Tukey's test (when a group is “ab”, this indicates that there are no statistically differences between “a” and “b”).

NA: not applicable.

^a Source: INFOR (2016).

^b This should be interpreted as the monthly interest rate (as a percentage) between the one-month choice and the six-month choice when participants change their short-term preference to the middle-term choice (six months).

^c This should be interpreted as the difference between the last safe option taken by participants before betting and a risk-neutral decision maker (the participant is indifferent between betting and not betting when the safe option is the same as the middle point between both coin sides). Therefore, larger positive numbers represent a greater risk aversion. See section in paper for further details.

the two religious groups. Both groups are quite similar, although plots owned by Catholic people are closer to urban areas. Evangelical decision makers have smaller pieces of land, and show a lower time discounting rate (more patience) and a higher risk aversion than Catholic decision makers, and they tend to trust more in people.

4.2. Risk aversion and time discounting

Tables 4 and 5 show the statistically significant variables related to risk aversion and time discounting, respectively. Trust was not a significant variable in these analyses, which indicates that the perception of the participants regarding the other people did not influence the results.

Variables that better predicted risk aversion were related to socio-economic and biophysical aspects. Higher schooling leads to lower risk aversion, as does ethnicity. In this case, *Mapuche* decision makers showed higher risk aversion than non-*Mapuche* ones. From a biophysical

¹⁰ In the case of risk aversion, 54 out of 160 answers corresponded to extreme risk aversion (people that always preferred the safest amount of money), while in the case of time discounting, consumption of forests and investment in tree plantations had a significant proportion of the answers as zero.

Table 3

T-Test for the main characteristics of the two religion groups.

Characteristic	Catholic (n = 93)		Evangelic (n = 39)	
	Mean	Std. dev.	Mean	Std. dev.
Decision maker's age (years)	63(a)	12	61(a)	13
Education (years)	7(a)	5	5(a)	4
Family size (quantity)	3.0(a)	2.0	3.0(a)	1.0
Distance to the closest town (km)	17(a)	11	25(b)	13
Distance to the largest city in the region (km)	68(a)	31	83(b)	35
Share of agricultural incomes (%)	49(a)	33	35(b)	31
Share of forestry incomes (%)	12(a)	19	12(a)	21
Share of off-farm incomes (%)	48(a)	29	59(a)	28
Total income (million Ch\$/year)	53(a)	179	43(a)	160
Formal land tenure (%)	74	NA	72	NA
Plot size (hectares)	130	297	79(b)	181
Forest cover (%)	36(a)	25	37(a)	29
Tree plantation cover (%)	5.2(a)	10.0	4.4(a)	6.2
Forest growth (m ³ /ha/year) ^a	5.2(a)	3.1	4.9(a)	3.0
Time discounting rate (%) ^b	7.4(a)	7.0	5.4(a)	4.2
Risk aversion rate ^c	7.3(a)	7.5	11.8	8.6
			(b)	
Trust question (% of people who trust)	54	NA	75	NA

Note: Letters (a and b) represent groups with statistically significant differences (95% confidence) according to t-Test (unequal sample size).

NA: not applicable.

^a Source: INFOR (2016).

^b This should be interpreted as the monthly interest rate (as a percentage) between the one-month choice and the six-month choice when participants change their short-term preference to the middle-term choice (six months).

^c This should be interpreted as the difference between the last safe option taken by participants before betting and a risk-neutral decision maker (the participant is indifferent between betting and not betting when the safe option is the same as the middle point between both coin sides). Therefore, larger positive numbers represent a greater risk aversion. See section in paper for further details.

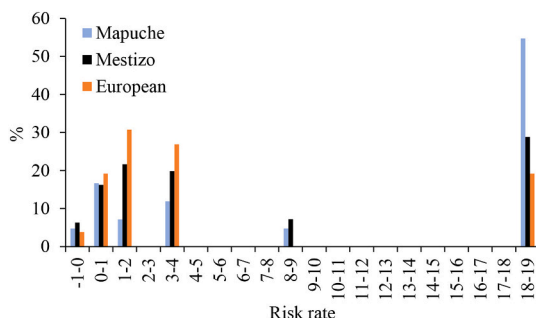


Fig. 3. Histogram of risk aversion. Mapuche (n = 42), Mestizo (n = 111) and European descendant (n = 26) decision makers.

Note: Negative risk rates indicate risk-prone decision makers. The last columns (risk rates between 18 and 19) express the maximum risk aversion rate, in which participants avoided betting until the safe amount of money was very low.

perspective, when the productivity of forests is higher risk aversion decreases. This indicates that people are willing to take more risks when their ecosystems are more productive.

In the case of time discounting, the most important explanatory variables were total income -which implies that richer people have lower time discounting rates (long-term preference)-, distance to the largest city in the region, and religion. As the distance to the city of Valdivia increases, time discounting rate decreases. A similar effect was observed with evangelical people (they tend to discount the future less than non-evangelical).

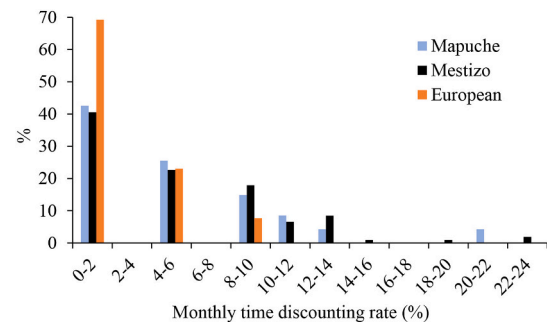


Fig. 4. Histogram of time discounting. Mapuche (n = 47), Mestizo (n = 106) and European descendant (n = 26) decision makers.

Note: Mapuche and Mestizos have a similar decay factor (reduction in the number of cases as the time discounting rate increases), while European descendants have a higher decay factor. Chile has low inflation (averaging 0.4% per month during 2014, which is very similar to the bank interest rate), so from a purely rational perspective, a high proportion of participants should be in the first time discounting group (between 0% and 2%). This is the case for the ethnic group of European descendants, in which almost 70% of participants have a monthly time discounting rate that is between 0% and 2%.

4.3. The use of forests

The decision maker's schooling¹¹ was the most important socioeconomic variable related to the intensity of timber extraction. After adjusting for schooling, timber extraction was also related to risk aversion. Risk aversion increases the intensity of logging, when decision makers at similar schooling levels are analyzed (Tables 6 and 7). Concerning tree plantation investments, income was the most important variable, and after adjusting by income time discounting was also significant. So, regardless of income, the time discounting rate is related to forest cover, with impatient/short term decision makers being less interested in planting trees.

5. Discussion and conclusions

5.1. Factors influencing risk aversion and time discounting

We evaluated the extent to which several socioeconomic and biophysical variables are related to risk preferences. As it was expected, schooling is one of the most important factors. Higher schooling reduces risk aversion (Rosen et al., 2003; Dohmen et al., 2010), which is relevant because education facilitates the adoption of new technologies (Knight et al., 2003) allowing a higher economic diversification (i.e. increasing of agricultural productivity).

The link between education and risk aversion is bidirectional. Education influences risk aversion (education as a cause), but it is also influenced by the risk aversion of parents (education as a consequence). Checchi et al. (2014) mention that risk aversion of parents negatively influences the education level of their children by reducing their educational expectations. Blackburn and Chivers (2013) observed a similar behavior when people have to invest in education using loans and other forms of indebtedness, which is usual in Chile. This situation can maintain higher levels of risk aversion over time, with negative consequences on forests as more years of formal education reduce economic dependency on natural resources, and prepare people to better face uncertainty.

By adjusting by schooling, which was highly correlated to income and plot size, the decision maker's ethnicity was also statistically significant concerning risk aversion. Mapuche decision makers showed

¹¹ This variable is correlated to total income. Both variables cannot be part of the same model, and schooling had a stronger effect.

Table 4

Result of the Tobit regression analysis for risk aversion.

Parameter estimates						
Parameter	DF	Estimate	Standard error	t Value	Pr > t	Variance inflation
Intercept	1	1.526617	0.146723	10.40	<0.0001	0
Ethnicity (Mapuche)	1	0.383428	0.123888	3.09	0.0020	1.11327
Education of the main decision maker (years)	1	-0.023684	0.009203	-2.57	0.0101	1.19671
Forest productivity (m ³ /ha/year)	1	-0.035017	0.016423	-2.13	0.0330	1.12095
_Sigma	1	0.544315	0.042061	12.94	<0.0001	0

Note: based on 151 cases.

Table 5

Result of Tobit regression analysis for time discounting.

Parameter estimates						
Parameter	DF	Estimate	Standard error	t Value	Pr > t	Variance inflation
Intercept	1	3.370053	0.490072	6.88	<0.0001	0
Logarithm of total income (\$)	1	-0.352610	0.068988	-5.11	<0.0001	1.03624
Distance to the largest city in the region (km)	1	-0.003451	0.001043	-3.31	0.0009	1.04817
Religion (Evangelic)	1	-0.157324	0.078896	-1.99	0.0461	1.06658
_Sigma	1	0.354919	0.031269	11.35	<0.0001	

Note: based on 142 cases.

Table 6

Result of the Tobit regression analysis for intensity of harvesting (consumption).

Parameter estimates						
Parameter	DF	Estimate	Standard error	t Value	Pr > t	Variance inflation
Intercept	1	0.941458	0.130935	7.19	<0.0001	0
Education of the main decision maker (years)	1	-0.034057	0.007062	-4.82	<0.0001	1.22804
Distance to the largest city in the region (km)	1	-0.002469	0.001064	-2.32	0.0203	1.08694
Logarithm of risk aversion	1	0.175723	0.081302	2.16	0.0307	1.18489
Forest cover (%)	1	-0.002637	0.001307	-2.02	0.0437	1.00416
_Sigma	1	0.414744	0.026364	15.73	<0.0001	

Note: based on 142 cases.

Table 7

Result of the Tobit regression analysis for tree plantation cover (investment).

Parameter estimates						
Parameter	DF	Estimate	Standard error	t Value	Pr > t	Variance inflation
Intercept	1	6.547211	3.233232	2.02	0.0429	0
Logarithm of forestry income (\$/year)	1	2.095066	0.387683	5.40	<0.0001	1.19735
Forest cover (%)	1	-0.149296	0.042505	-3.51	0.0004	1.00160
Logarithm of time discounting rate	1	-9.728376	3.605127	-2.70	0.0070	1.19890
_Sigma	1	10.902633	0.933855	11.67	<0.0001	

Note: based on 145 cases.

higher levels of risk aversion¹² than non-Mapuche decision makers, at similar socioeconomic level. This result contrasts with findings by [Henrich and McElreath \(2002\)](#) in previous research in Chile. While their experiments involved real payments to participants, which could be behind this discrepancy, there are more fundamental differences between both studies. They compared landowners (rural Mapuche) and non-landowners (urban Mestizos) in a region dominated by Mapuche people (Chol-Chol in the Araucanía Region), while this research compared decision makers in a territory that is not dominated by Mapuche. [Yesuf and Bluffstone \(2009\)](#), [Guiso et al. \(2014\)](#) and [Buccioli and Zarri \(2015\)](#) note that there is a significant correlation between previous experiences and risk aversion, and when past experiences are negative, people become more risk-averse. This could be the case for the Mapuche

decision makers living among Mestizos and European descendants, because they have experienced more discrimination than other ethnic groups. Therefore, the positive relationship between risk aversion and Mapuche ethnicity could result from the social context of Mapuche decision makers rather than the Mapuche ethnic group itself. In other words, the variable for ethnicity in our study is less a measure of ethnicity itself as it is a measure of ethnically based discrimination that has taken place over time.

In the case of time discounting, income and religion were significant variables in the analysis. In the case of income, when it increases people turn more patient (time discounting rate decreases). [Lawrance \(1991\)](#), [Yesuf and Bluffstone \(2009\)](#) and [Tanaka et al. \(2010\)](#) observed the same relationship between income and time discounting rate, while [Becker and Mulligan \(1997\)](#) demonstrated that “wealth causes patience”, and [Dohmen et al. \(2015\)](#) that “patience causes wealth”. Therefore, the relationship between income and time preference is mutually reinforced, and it is not possible to know in this case if income is a cause or a

¹² In a context of uncertainty, risk-averse decision makers prefer safer decisions, although payoff is lower.

consequence of time discounting rate.

After adjusting by income, Evangelical decision makers had lower time discounting rates (more patience) than non-Evangelical ones, at the same income level (Table 3). Evangelical decision makers were shown to be more patient than Catholic ones (they better tolerate uncertainty related to time). This is consistent with that of Carter et al. (2012), who show that religion results in a higher valuation of the future. Similarly, Paglieri et al. (2013) observed that Protestants (Dutch Calvinists) were more patient than Catholics. McCullough and Willoughby (2009) note that religion is related to self-control and self-regulation (patience), especially those religions that promote intense meditation, prayer, readings, and other rituals, which is the case of the Evangelical church in Chile. A similar finding was observed by Liebenehm and Waibel (2014) in West Africa, where landowners who spent more time in a religious school were less risk averse and more patient than others. It is important to note that almost half of Mapuche decision makers were Evangelical, so that the interaction between ethnicity and religion would be also producing different behaviors concerning uncertainty.

5.2. Factors influencing the use of forests

According to the models, and after adjusting by schooling, larger risk aversion levels relate to higher intensities of timber extraction, which can lead to degrade forests. Bonin et al. (2007) mention that more risk-averse decision makers tend to have occupations with lower earning risk, because people prioritize financial safety over profit. This is the case of logging forests in Chile, since this activity is usually less profitable than cattle raising or agriculture (Reyes, 2020), but safer from an economic perspective (because the firewood market is very stable). So, for an important share of decision makers, uncertainty means harvesting forests more intensively.

Additionally, there is a close relationship between risk aversion, forest productivity,¹³ and the intensity of timber extraction. As it was earlier mentioned, a larger risk aversion can result in a higher intensity of timber extraction, which in turn can reduce forest productivity along time by degrading forests. At the same time, a reduced forest productivity can increase risk aversion. A similar circular relationship was observed by Tanaka and Munro (2014) in Uganda, who mention that decision makers living in regions with low-productivity show greater risk aversion. It is difficult to know if forest productivity is a cause of risk aversion, a consequence, or both, as could be the case of decision makers living in chronic poverty (poverty traps) (Yesuf and Bluffstone, 2009). Risk-averse decision makers tend to opt for low-risk and lower-return activities, such as timber extraction (the checking account). According to this, if a less productive ecological context coincides with low-education decision makers, the probability of a chronic degradation on forests would be higher. This could create a self-reinforcing dynamic between social marginalization and degraded ecosystems in an ongoing vicious cycle over time (degradation loops) (Dercon and Christiaensen, 2011).

Moreover, the strong and mutually reinforced relationship between income and time discounting rate could produce positive contexts for forest conservation, as the future consumption acquires value (forest as an investment and saving account). This is confirmed in this research, since a lower time discounting rate (more patient people) was related to a higher tree plantation cover (%). In other words, more and better income opportunities for decision makers have positive implications for forests, not only because people have less need to cut trees (Reyes et al., 2018), but also because they change their perception of uncertainty related to time and are more willing to accumulate value in their forests (investing in forest recovering, and increasing forest cover).

5.3. Ethnicity, religion, and forest policy implications

These results are indicative that ethnicity and religion are correlated to differences in attitudes about risk and time. They determine different choices in how forests are used. However, **there is not a direct relationship between ethnicity and religion and the use of forests, as was hypothesized**. Indeed, this research tried to link these cultural characteristics to the use of forests but found no relationships. Therefore, there are not ethnic and religious groups that are more or less able to manage their forests in a sustainable way, but specific socioecological contexts that determine differences in the way people face uncertainty, by changing their attitude concerning risk and time, which can ultimately influence forests.

The socioecological context where decision makers work is strongly influenced by institutions and policies. Unlike other developing countries, where informal land tenure is one of the main sources of uncertainty about the use of natural resources (Kissinger et al., 2012), in Chile uncertainty has been more exacerbated by poorly designed policies. The Forest Management Plan (FMP) and related subsidies, which is mandatory for all decision makers who want to harvest their forests, has several characteristics that increase forest owners' uncertainty: a) the FMP is difficult to understand due to its high technical complexity, so people harvest their forests without understanding the document that was approved by the National Forest Service (CONAF); b) CONAF focuses its resources in controlling approved FMP, instead of overseeing illegal loggers. Therefore, people managing forests legally (but without understanding the FMP) have a higher probability of being controlled, and therefore a higher probability to be fined if there was some failure to follow the FMP; c) in order to get subsidies to manage forests, decision makers have to compete for resources (which has a cost); and c) subsidies to manage forests are paid between two and five years after silvicultural activities are implemented (so people have to be patient), and money is not reimbursed if something goes wrong (and risk tolerant).

For risk-averse decision makers the FMP represents a risky bet, and many of them would not be willing to use it. That is why, after 40 years of the enactment of the regulation that encourages the use of Forest Management Plans (FMP), only a small share of decision makers uses this legal tool. This is especially true in the case of small and medium landowners, who have lower schooling, people whose incomes strongly depend on forest production, marginalized decision makers due to socioeconomic reasons, racism, etc. Unfortunately, they are who harvest their forests in a more intensive way, so where the FMP (one that can be understandable for anybody) would be more necessary.

In Chile, managing uncertainty related to the use of forests should be especially careful in contexts with high poverty levels, social conflicts and low productivity ecosystems, like those in the Southern regions, in order to avoid entering in degradation loops. This is, for example, the case of the Chiloé Island, in which forest degradation has been reported in the last decades (Bannister, 2018). In this case, uncertainty about the use of forests has reduced the resilience of the socioecological system and augmented its vulnerability (Berkes, 2007). In other regions of Chile, an improvement in people's income has resulted in a higher investment on tree plantations, which are being used to produce firewood, especially from *Eucalyptus* sp. This can reduce the pressure on native forests (Paquette and Messier, 2010; Reyes, 2017).

In this context, the reasons as to why forest policies in Chile have been ineffective in addressing forest management (Lara et al., 2016), such as those reported above, should be seriously addressed in order to promote changes towards true management of these forests. We argue greater attention should be paid to reduce the uncertainty associated with forest regulations. These could be done by adapting regulations to be more reflective of decision makers' needs, pace and knowledge, for instance by adapting the technical language to the decision makers' language. In sum, what is really important for promoting the sustainable forest management in Chile is having a regulatory body that generates

¹³ In this case, forest productivity depends on the state of forests.

certainties, because that is going to create a better environment for decision making concerning forests, by reducing the intensity of timber extraction and increasing the willingness of planting trees for the future.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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